

L#01: Units, Charge, Current, Voltage

2018-02-05

Six basic SI units – International system of unit

	Quantity	Basic unit	Symbol
1.	Length	meter	m
2.	Mass	kilogram	kg
3.	Time	second	s
4.	Electric current	Ampere	A
5.	Thermodynamic temperature	Kelvin	K
6.	Luminous intensity	<u>candela</u>	cd

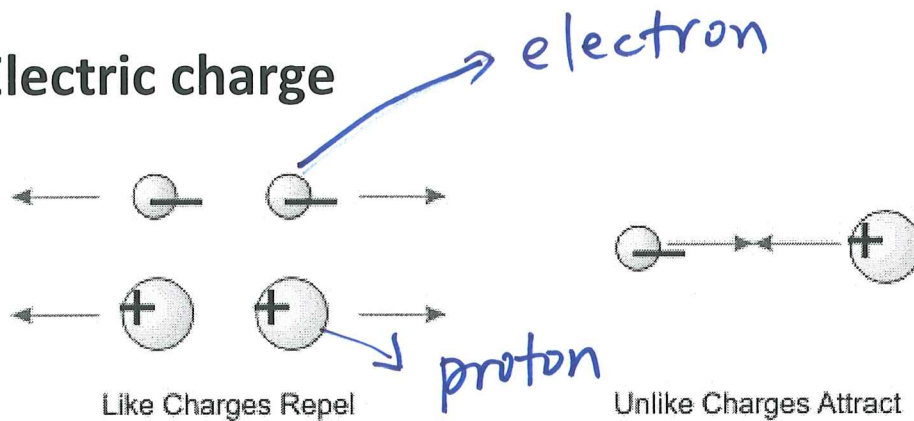
Derived units in circuit analysis

Quantity	Unit	Symbol
Electrical charge	Coulomb	C
Electric potential	Voltage	V
✓ Resistance	ohm	Ω
Conductance	Siemens	S
Inductance	henry	H
Capacitance	farad	F
Frequency	Hertz	Hz
Force	newton	N
Energy, work	joule	J
Power	watt	W
Magnetic flux	Weber	Wb
Magnetic flux density	Tesla	T

Decimal multiples and submultiples

Factor	Prefix	Symbol
<u>10^9</u>	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ ←
10^{-9}	nano	n
10^{-12}	pico	p

Electric charge



- Charge is an electric property of the atomic particles, measured in coulomb (C).
- The charge e on one electron is negative and equal in magnitude to $1.602 \times 10^{-19} \text{ C}$.
- **Law of conservation of Charge:** Charge can neither be created nor destroyed, it can only be transferred.

Question:

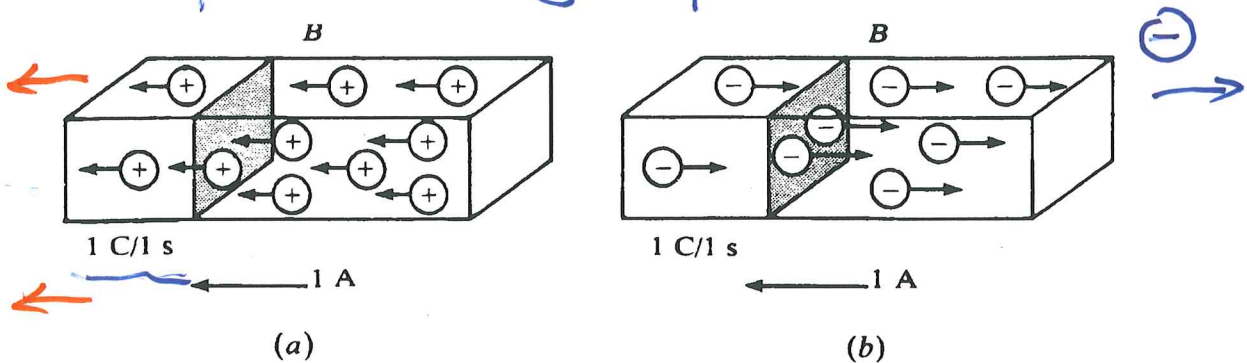
How many electrons in one Coulomb?

$$1.602 \times 10^{-19} \text{ C} \times x = 1 \text{ C}$$

$$6.24 \times 10^{18}$$

Current

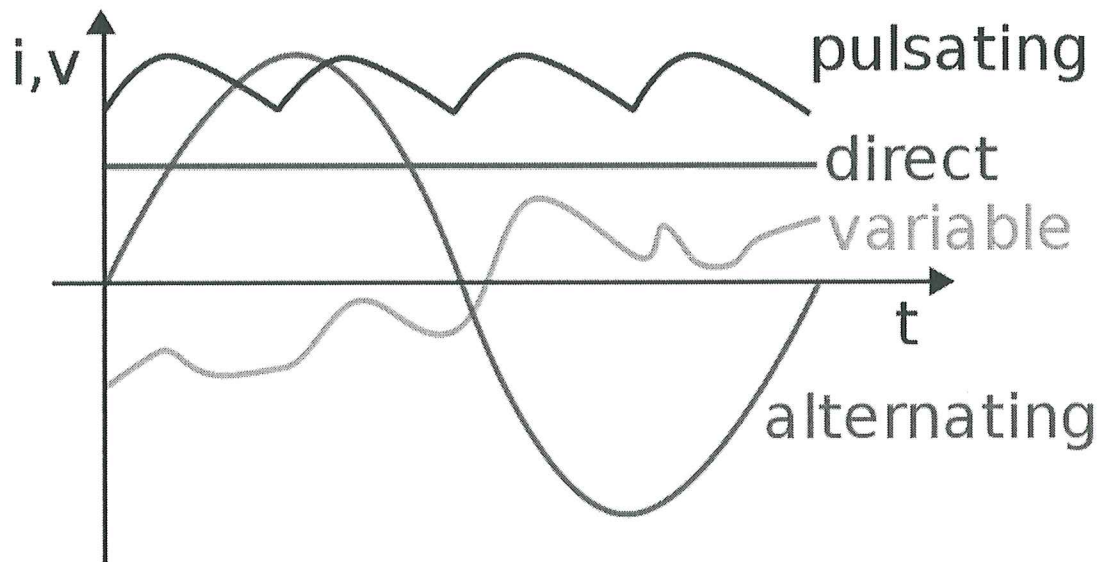
- **Electrical Current:** motion of charge creates current.
- **Convention:** Current flows in the direction of positive charge : protons.



- Current is measured in Ampere (A), which is 1 C per second.
- Formula:

$$i = \frac{dq}{dt} \quad (A)$$

AC & DC



Source: https://en.wikipedia.org/wiki/Direct_current#/media/File:Types_of_current.svg

Direct current (DC): the electric current flows in a constant direction

Alternating current (AC): the electric current periodically reverses directions

Q1: Calculate the amount of charge represented by four million protons.

Q2: The total charge entering a terminal is given by $q = (10 - 10e^{-2t})$ mC. Find the **current** at $t = 0.5$ s.

$$i(t) = \frac{dq(t)}{dt}$$
$$= \frac{d(10 - 10e^{-2t})}{dt}$$
$$= 0 - (-2) \times 10e^{-2t}$$

$$i(t) = 20e^{-2t}$$

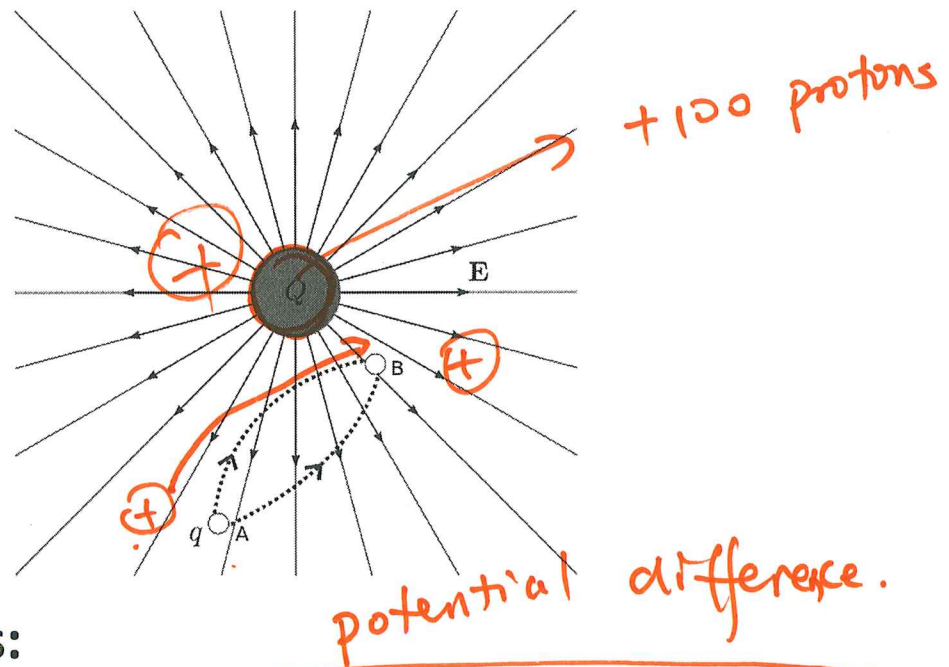
$$i(0.5) = 20e^{-2 \times 0.5} = 20e^{-1}$$

$$= 7.4$$

mA

Q3: Determine the total charge entering a terminal between $t = 1$ s and $t = 2$ s if the current passing the terminal is:
 $i = (3t^2 - t) \text{ A}$

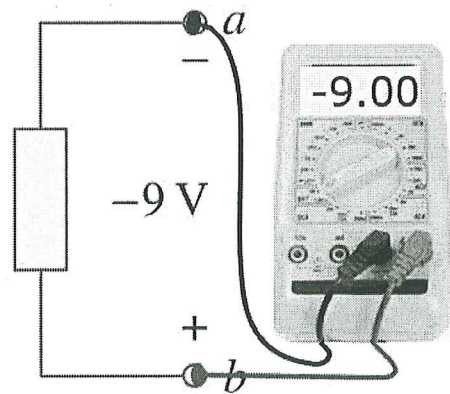
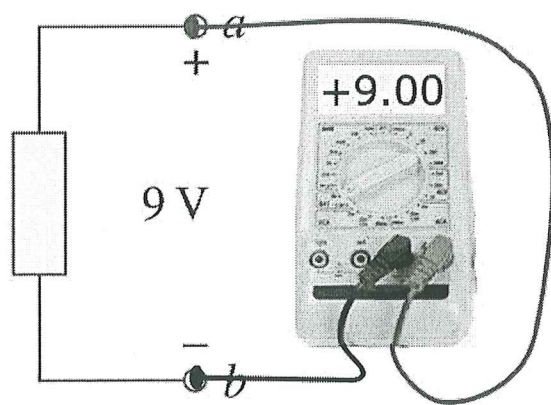
Voltage



Definitions:

- Voltage is the difference in electric potential between two points in an electric field.
- The potential difference is the work done per unit of charge against a static electric field to move a test charge between two points, which is measured in volts (a joule per coulomb).

$$\underline{V_{ab}} = \frac{dw}{dC} = \boxed{V_{a0}} \downarrow V_{b0}$$



$V_{ab} > 0$	$\rightarrow V_a > V_b$
$V_{ab} < 0$	$\rightarrow V_a < V_b$
$V_{ab} = 0$	$\rightarrow V_a = V_b$
$V_{ab} = V_a - V_b = -V_{ba}$ $= -(V_b - V_a)$ $= V_a - V_b.$	